# A. Easy Peasy Lemon Squeezy

Author: ***Shahril***

This is a very easy question, indeed.

## Input

The first line contains $T$ $(1 \le T \le 100)$, the amount of test case(s).
The following T lines contains **A, B** $(1 \le A < B \le 5000)$ and **S** $(1 \le S \le B\text{-}A)$ integers.

## Output

Output the numbers sequence between A and B (inclusive), with S step(s), with the number sequence should be separated by a single space.
Each answers should be separated by a newline.

| **Sample Input** | **Sample Output** |
|---|---|
| 3<br>1 2 1<br>1 10 2<br>99 100 2 | 1 2<br>1 3 5 7 9<br>99 |

# B. Credentials Check

Author: ***Shahril***

Imagine a system with a set of credentials which contain username and password. Can you make a simple system that check if user entered credential is match with the one inside the list?

## Input

First line contains $C\ (5 \le C \le 100)$, number of credentials.

For the following $C$ lines there will be two strings, U and P, representing username and password in the system. U and P will contain no space in between them.

Next line will contains $T\ (1 \le T \le 100)$, number of test case(s).

The following T lines will contain two test credential, UT and PT, representing username and password that we want to check. UT and PT will contain no space in between them.

## Output

For each T lines, print "YES" if the credential exists inside the list, otherwise print "NO".

| **Sample Input** | **Sample Output** |
|---|---|
| 4<br>admin admin<br>john abc123<br>surya password<br>test test<br>5<br>abu bakar<br>john abc123<br>user pass<br>sura password<br>surya password | NO<br>YES<br>NO<br>NO<br>YES |

# C. Alien Communicator

Author: ***Harith***

An alien just landed on Planet Earth.
He can speak English but every sentence he talked and heard is backwards.
Create a program to communicate with him.

## Input

First line contains **T (1 ≤ T ≤ 10)**, the number of test case(s).
For the next T lines, it will contain a single English statement.

## Output

For each sentences in the statement, print the sentences backward.

| **Sample Input** | **Sample Output** |
|---|---|
| 3<br>good morning<br>how are you<br>do you come in peace | morning good<br>you are how<br>peace in come you do |

# D. Local Extrema

**Author: *Ikhwan***

You are given list *Z.* Elements of this array $Z_i$ will be consider as *local minima* if the element is strictly less than both of its neighbors ($Z_i < Z_{i-1}$ and $Z_i < Z_{i+1}$). Also the element will called *local maxima* if the element is strictly greater than its neighbors ($Z_i > Z_{i-1}$ and $Z_i > Z_{i+1}$). Since $Z_1$ and $Z_n$ have only one neighbors each, they are neither *local minima* nor *local maxima*.

An elements is called *local extrema* if it either *local minima* or *local maxima.* Your task is to calculate the number of local extrema in the array.

# Input

The first line of the input contains one integer *n* ( $1 \leq 1000$ ) which denotes the number of element in the array.

The second line will contains *n* integer $Z_1, Z_2, .., Z_n$ ($1 \leq Z_n \leq 1000$) which denotes the elements of array *Z*

# Output

Print the number of local extrema in the array

| Sample Input | Sample Output |
|---|---|
| 5<br>2<br>3<br>1<br>2<br>5 | Local Exima : 2 |
| 3<br>12<br>34<br>33 | Local Exima : 1 |

# E. Counting

Author: ***Shahril***

Give a statement, count how many time the word occurrences inside the string.

## Input

The first line contains S, an English statement.
The next line contains $T\ (1 \le T \le 100)$, the amount of test case(s).
The following T lines a word that you need to find the occurrences.

## Output

Output the number word occurrences inside the string.

| **Sample Input** | **Sample Output** |
|---|---|
| hello hello abc hi ais kacang hi<br>3<br>hello<br>hi<br>kacang | 2<br>2<br>1 |

# F. Alphabet & Number Occurrences

Author: ***Fahmi***

Write a program that accepts a sentence. Then you will need to count the number of letters and digits in the sentence.

**Input**

The first line contains ***T (1 ≤ T ≤ 10)***, the amount of test case(s).
The following T lines a single statement, S.

**Output**

Print number of alphabets and digits in each sentences, using format "Alphabet: x, Digit: x"

| **Sample Input** | **Sample Output** |
|---|---|
| 2<br>Hello Python 3333<br>Abc123 | Alphabet: 11, Digit: 4<br>Alphabet: 3, Digit: 3 |

# G. Discount

Author: ***Shahril***

Some small supermarket is having a sales promotion. If you buy exceeds some threshold value you will get a discount. The discounts are based on the table below:

| Value | Discount |
|---|---|
| ≤ RM500 | 0% |
| RM501 - RM1000 | 5% |
| RM1001 - RM2000 | 8% |
| > RM2000 | 15% |

Harith want to shop, but then he needs to plan carefully how much we want to buy. So please help him calculate the final amount based on total price.

## Input

The first line contains $T\ (1 \le T \le 100)$, the amount of test case(s).
The following T lines contains **p** $(0 \le P \le 100000)$, represent the total price Harith planned to buy.

## Output

Output the final calculated price, based on the table above. Price must be rounded down to two decimal places.

| Sample Input | Sample Output |
|---|---|
| 5<br>0<br>100<br>1000<br>1337<br>5000 | RM 0.00<br>RM 100.00<br>RM 950.00<br>RM 1230.04<br>RM 4250.00 |

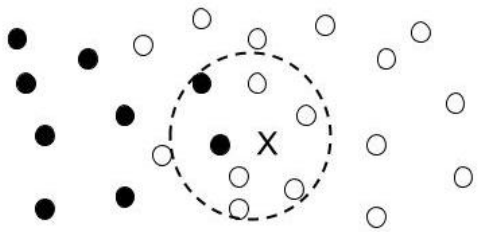
# ***k*-NN**

Author: ***Shahril***

In *Artificial Intelligence*, there is a subset field called ***machine learning*** which provides the computer the ability to automatically learn and improve from experience without being explicitly programmed.

One of such simple and very popular technique is ***k*-NN (K-Nearest Neighbours)**.

*k*-NN works by searching the ***k*** nearest points (neighbours) to the desired point. By getting the nearest points, *k*-NN will try to make an assumption based on the nearest points' classes and use that to predict the class of the desired point. In A.I. this is also called classifying problem as we want to predict the class of the desired point.




***k*** = 7

For example, above is set of points with two classes (***A = black circle***, ***B = transparent circle***). ***X*** is a point that we desired to predict. *k* is number of nearest points that we want to find (which in this case is **7**). By searching the distances between X and all points, we then sort out the result. Based on the *nearest seven points* to the X, we found out that there are **2 and in class A and 5 are in the class B.** By getting this information, k-NN will choose the highest frequency among all the classes and uses that determine class of X (which in case **X = class B**).

## Input

First line of input has ***P (5 ≤ P ≤ 20)***, which represent number of points in the cartesian plane. For each P lines, there will contain three input X, Y, C where the first two input represent the coordinate of the point and C represent *its class*.
***X (0 ≤ X ≤ 10), Y (0 ≤ Y ≤ 10), C (1 ≤ C ≤ 2).***

Next line will contain ***T (1 ≤ P ≤ 5)***, where it represents the number of test case(s). For each next T lines, there will be three value I, J and K, where I and J represent location of point where we want to predict, and K represent how much nearest nodes we want to compare.
***I (0 ≤ I ≤ 10), J (0 ≤ J ≤ 10), K (1 ≤ K ≤ P) and K is odd number.***

## Output

Print the the case number 'Case #n:'; n is test case number followed by class predicted by your k-NN algorithm. Please note that there will be only two classes, 1 and 2.

| **Sample Input** | **Sample Output** |
|---|---|
| 5<br>4 2 1<br>7 2 2<br>3 6 1<br>7 4 2<br>2 3 1<br>3<br>3 4 1<br>5 4 3<br>4 4 5 | Case #1: 1<br>Case #2: 2<br>Case #3: 1 |

Hint:

$$\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

# I. Search and Destt..Replace!

Author: ***Shahril***

Y'all have heard about search and destroy, right?
Well this question will require you to search and replace instead!!

## Input

The first line contains original text, T.

The next line contains ***N (1 ≤ N ≤ 100)***, amount of character that you need to search and replace.

The next N lines will contain two single characters, S and R, which S representing character that you need to find, and R representing character that you need to replace. Please note that S and R can be any characters except for space " ".

## Output

Output the final string.

| **Sample Input** | **Sample Output** |
|---|---|
| Y'all have heard about search and destroy, right?<br>5<br>? !<br>a 4<br>i l<br>o @<br>h $ | Y'4ll $4ve $e4rd 4b@ut se4rc$ 4nd destr@y, rlg$t! |

# J. Basic Statistics

Author: ***Shahril***

Easy question! Please answer this one :-(

## Input

Sequence of numbers with odd length.

## Output

Mean and Median of the sequence.
The answer must be rounded to two decimal places.

| Sample Input | Sample Output |
|---|---|
| 12 12 -63 23 -91 -49 -65 -32 12 | Mean : -26.78<br>Median : -91.00 |

# K. Uppercase Title

Author: ***Shahril***

Given a statement, convert all first letter of every words into capital letter.

## Input

The first line contains $T\ (1 \le T \le 100)$, the amount of test case(s).
The following T lines contains string S, which is going to be converted.

## Output

Converted string with all first letter of every word capitalized.

| **Sample Input** | **Sample Output** |
|---|---|
| 5<br>a b c<br>aa bbb cccc<br>linus torvalds<br>welcome to prosolve 4.0!!!<br>muhammad ibn musa al Khwarizmi | A B C<br>Aa Bbb Cccc<br>Linus Torvalds<br>Welcome To Prosolve 4.0!!!<br>Muhammad Ibn Musa Al Khwarizmi |